INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

iOS Forensics

Case 2 iPhone SE iOS 13.4.1

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SBT-DF205 Mobile and IoT Forensics Case Study

Module 2 iOS Forensics

In this training case, Harbour Operations Ltd requests a forensic review of a company-issued iPhone SE following a dispute about messages, shared media and the device's movements. As the examiner, establish the associated accounts and reconstruct relevant activity. Assess whether communications, browsing, photographs and device records support a consistent account of events.

Use the original identities and historical dates in the supplied dataset. Base your conclusions on the records examined and explain any findings that remain uncertain.

mv old_iOS_13 new_ SBT-DF205-C2

Evidence and tools

Use Joshua Hickman's iOS 13.4.1 evidence package hosted by Digital Corpora. Download ios_13_4_1.zip, extract it, then extract the file system archive at Extraction/13-4-1.tar inside the package. Use that file system for the graded tasks.

Download the iOS 13.4.1 evidence package

The download is approximately 9.63 GB. Allow at least 60 GB of free disk space. Work in a Linux VM or macOS to preserve Unix paths and links. Use an archive extractor, DB Browser for SQLite or SQLite, a binary/XML plist reader, ExifTool and a spreadsheet editor. Refer to the supplied iOS slides for examination procedures.

Examination instructions

- Preserve the source ZIP and TAR. Examine working copies and keep matching database -wal and -shm files together.
- Work only on the supplied extraction. Jailbreaking, resetting devices and accessing live accounts are outside this activity.
- Record source paths and record IDs. Treat /var and /private/var aliases carefully to avoid counting the same evidence twice.

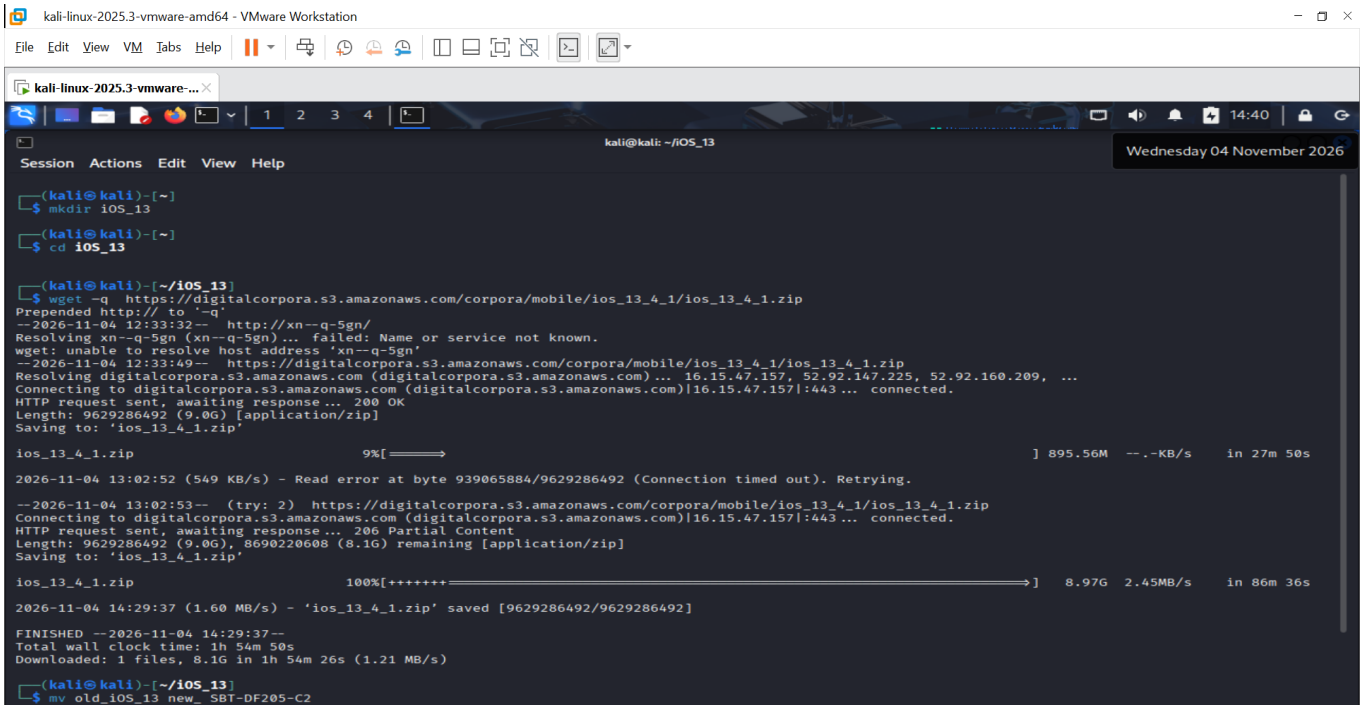
- Retain raw timestamps, normalise the timeline to UTC and document empty or unavailable records. Exclude passwords, tokens and unrelated personal data from your report.

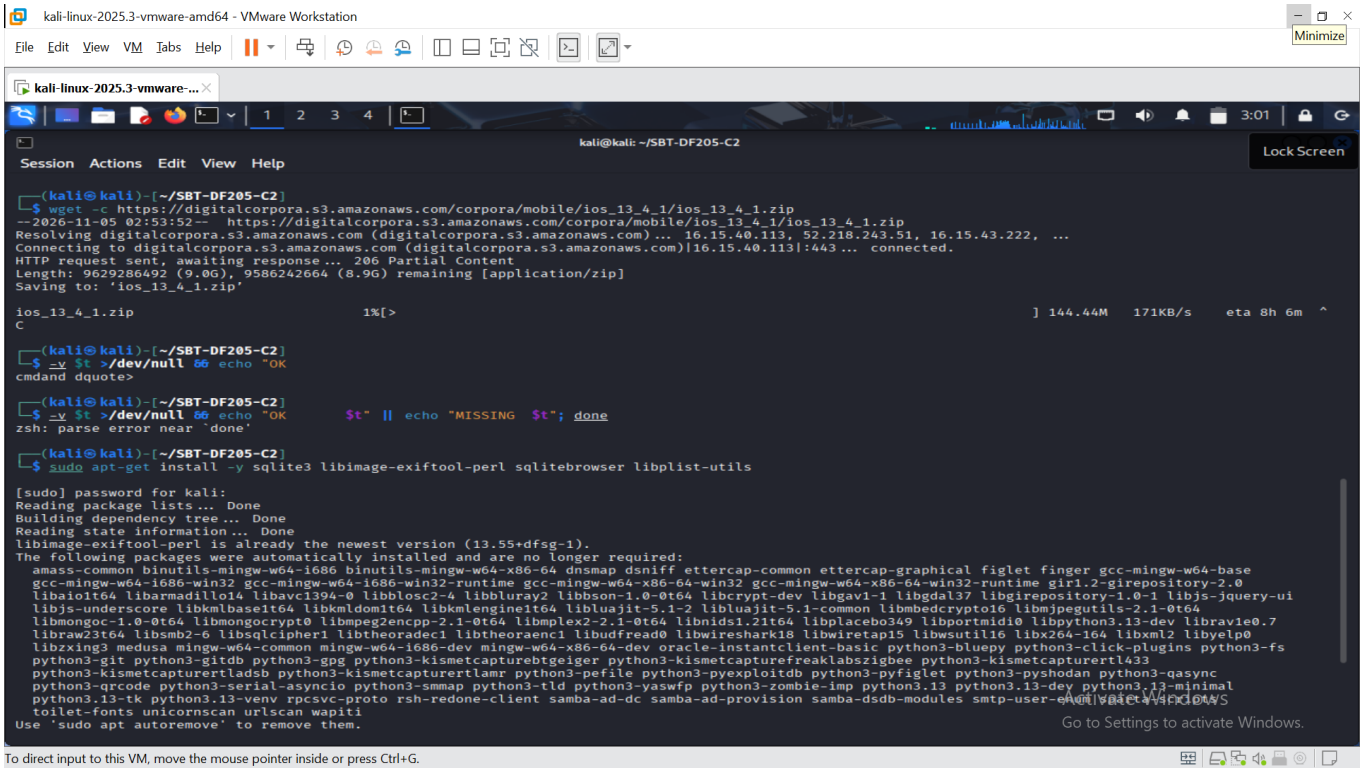
Practical investigation tasks

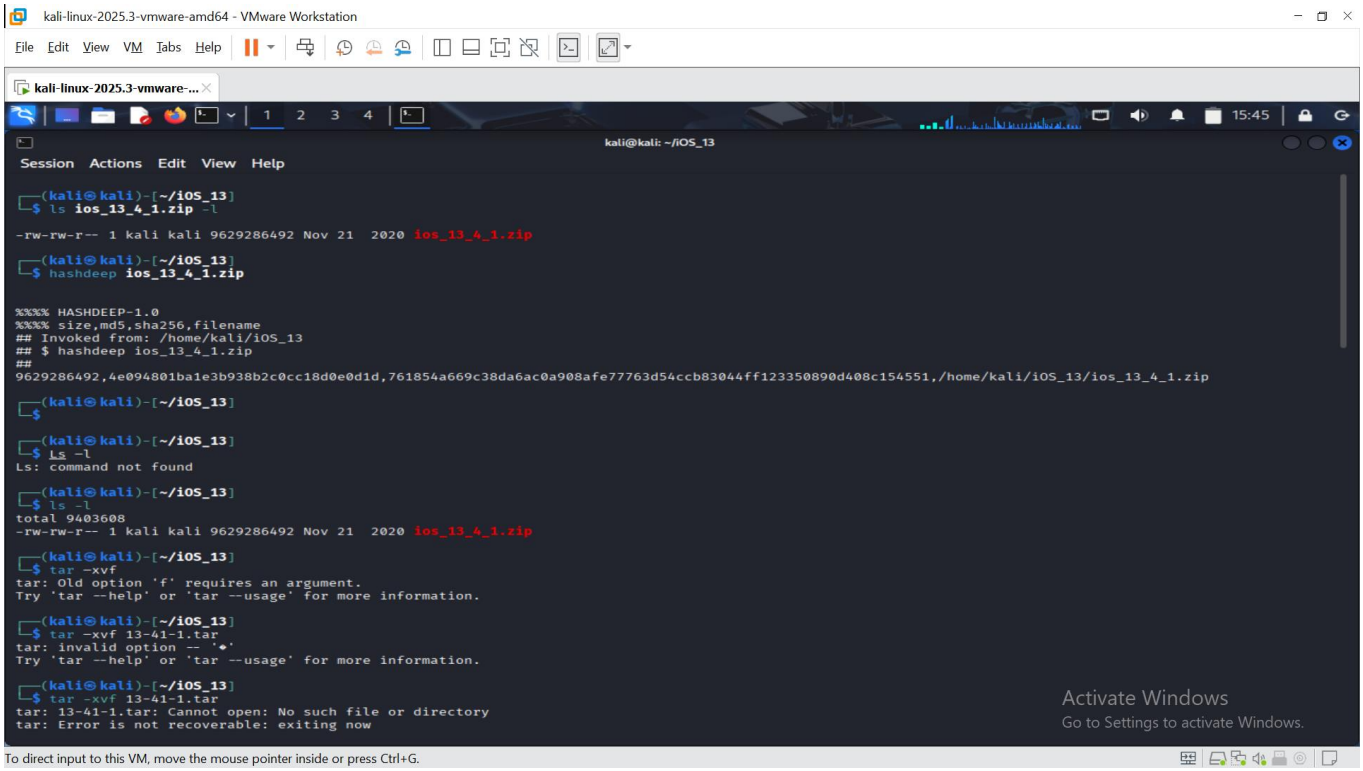
Task 1 Evidence preservation 10 marks

1. Create a handling log using case reference SBT-DF205-C2. Record the evidence source, archive names, sizes, receipt time, examiner and tool versions.

2026-11-04 14:29:37 (1.60 MB/s) - 'ios_13_4_1.zip' saved [9629286492/9629286492]







2. Calculate SHA-256 values for the source ZIP and TAR before examination and recheck them at completion. Compare with a trusted manifest for the exact files if supplied; otherwise record your values as baselines.

43043828,d65dc98ff6e7d0c7afac041daf9fb094,541ad51e24e0c47b4ea9e6607687308c2a2406f81be0f6dd894ec467c2e394ad,/home/kali/SBT-DF205-C2/ios_13_4_1.zip

hashdeep ios_13_4_1.zip:
151459577,76876c895039efe3a4f12509bec4a55c,7b8a59ce6c870fdd28b2da5f91a6176f102d287bbdac98be2e4a0a175bff7bb7,/home/kali/SBT-DF205-C2/ios_13_4_1.zip

MD5: **76876c895039efe3a4f12509bec4a55c ios_13_4_1.zip**

SHA-256: 7**b8a59ce6c870fdd28b2da5f91a6176f102d287bbdac98be2e4a0a175bff7bb7 ios_13_4_1.zip**

TAR:

TAR

3. Maintain an evidence register for selected files, including evidence ID, source path, size and SHA-256. Record extraction errors, processing actions and database sidecars.

Task 2 Device accounts and settings 10 marks

1. Establish the device model, iOS version and build from configuration or log artefacts. Cite the exact source and property keys.
2. Examine Accounts3.sqlite and relevant plists to identify account associations. Corroborate one association using a second source.

3. Report two supported settings, such as location services, backup information or effective user settings. Distinguish stored configuration from proof of user activity.

Task 3 Messages contacts and calls 20 marks

1. Locate private/var/mobile/Library/SMS/sms.db and inspect its schema. Extract at least five messages and reconstruct one conversation, showing record IDs, participants, content, direction, service type and raw and converted timestamps.

2. Trace one message attachment record to its local file where available. Explain the relationship and any unavailable content. Validate status meanings against the schema and application version.

3. Use AddressBook.sqlitedb to associate at least two message or call identifiers with stored contacts where matches exist. Examine at least three calls in CallHistory.storedata, including direction, time and duration.

Task 4 Safari and calendar 10 marks

1. Examine Safari History.db and export at least five visits with URLs, titles, raw times and UTC conversions. Distinguish recorded visits from bookmarks, open tabs and synced tabs.

2. Inspect Calendar.sqlitedb and Extras.db where present. Record relevant tables and row counts, then report any supported event details. Document empty tables without inventing appointments.

Media activity and location analysis

Task 5 Photographs and media 15 marks

1. Examine Photos.sqlite, including ZGENERICASSET and related metadata tables as present. Select at least three media records and locate their corresponding files where available.

2. Record filenames, paths, hashes, capture-related and added times, available offsets and GPS metadata. Distinguish media captured by the camera from imported or saved media using supporting fields.

3. Validate one database result against file metadata. Explain discrepancies and any supported sharing or deletion indicators without assuming an indicator proves a complete transfer or deliberate deletion.

Task 6 KnowledgeC activity 10 marks

1. Locate private/var/mobile/Library/CoreDuet/Knowledge/knowledgeC.db. Inspect ZOBJECT and related metadata, then extract at least five relevant records, such as application focus or device lock activity.

2. Report the stream name, record ID, associated application or value, raw start and end times, UTC conversions and duration where meaningful. Correlate at least one interval with messages, browsing or media activity.

Task 7 Health and location evidence 15 marks

1. Inspect healthdb.sqlite and healthdb_secure.sqlite where present. Identify a recorded data source or device and examine at least two activity samples. Report times, values and units without treating a synced wearable record as proof the phone holder performed the activity.

2. Inspect available routined Local.sqlite and locationd cache_encryptedB.db records. Examine at least three location-related entries where available, recording source tables, coordinates, times and their meanings.

3. Distinguish learned visits from Wi-Fi or cell reference locations. Use supported visit start and end fields where available; a record creation time is not automatically a visit time. Compare one location result with media or activity evidence.

Task 8 Timeline and conclusions 10 marks

1. Create a timeline of at least 12 distinct events from four or more artefact families. Include raw time, field name, epoch and unit, UTC conversion, event description and evidence ID.

2. Present at least three correlations and explain two alternative interpretations. State what the evidence establishes, what remains uncertain and what further evidence would resolve the gaps.

Time handling. Many Apple dates use a 1 January 2001 epoch, but units vary by field. Validate each conversion. The creator's activity notes use UTC-04:00. Keep media times without a known offset explicitly unresolved.

Submission and marking

Submit one ZIP file named SBT-DF205_Case2_fwsd2511478@students.icdfa.edu.ngr.zip through E-Campus. Replace RegistrationNumber with your identifier, using hyphens instead of slashes. Include the following items.

Submission item	Required content
Report.pdf	A concise forensic report covering scope, method, findings, timeline, conclusions and limitations. Normally 4 to 6 pages, excluding appendices.
Evidence_Register.csv	Evidence IDs, source paths, record references, file sizes and SHA-256 values.
Timeline.csv	At least 12 distinct events with raw times, validated UTC conversions and evidence references.
Queries.sql	SQL used, with the target database and purpose identified.
Handling_Log.csv	Receipt and processing actions, tool versions, errors and integrity checks.
Supporting evidence	8 to 12 legible, captioned screenshots and selected metadata or plist output.

Include only the extracts needed to support your findings. Do not submit the full evidence archives, entire databases or live account credentials. Cite the supplied slides and any technical sources used.

Marking rubric

Task	Assessment focus	Marks
1	Evidence handling, hashes and register	10
2	Device identification, accounts and supported settings	10
3	Messages, attachment tracing, contacts and calls	20
4	Safari records and calendar examination	10
5	Media analysis, metadata and interpretation	15
6	KnowledgeC records and activity correlation	10
7	Health data sources and careful location analysis	15
8	Timeline, correlations and justified conclusions	10
Total		100

Marks depend on accurate, reproducible findings and clear source references. A documented absence or inconclusive result can earn credit when the examination method and limitations are explained. Submit your own examination, supported by the underlying records.

Evidence attribution: Joshua Hickman, iOS 13.4.1 iPhone SE dataset, Digital Corpora. Course reference: the supplied iOS forensic investigation slides.